NEUROMORPHIC COMPUTING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Neuromorphic Computing

Question 1

10.0 points

Examine the diagram below of a leaky integrate-and-fire (LIF) neuron model, which shows the membrane potential trace over time. Analyze how the membrane potential changes in response to incoming spikes and explain the conditions under which the neuron fires. Specify any key parameters illustrated in the diagram, such as threshold potential and leakage rate.

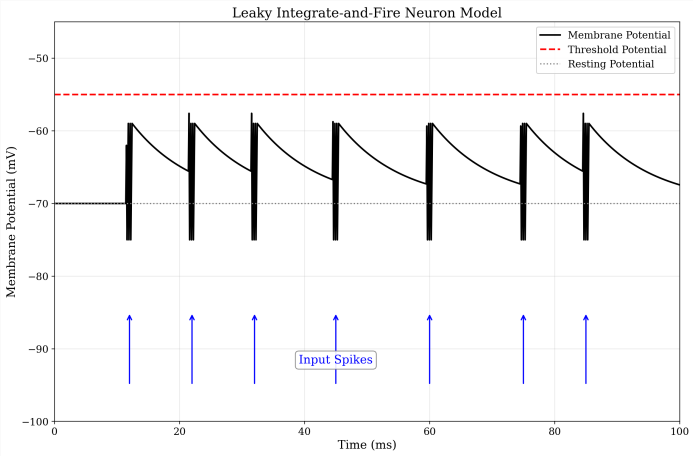

Fig. 1. A membrane potential trace of a leaky integrate-and-fire (LIF) neuron model. The diagram shows how the membrane potential changes over time in response to incoming spikes. Key parameters such as the threshold potential and leakage rate are illustrated. The trace should depict the membrane potential rising with incoming spikes, decaying due to leakage, and spiking when the threshold is reached.

Question 2 10.0 points

Examine the diagram below illustrating a spike-timing dependent plasticity (STDP) learning rule. Describe how the timing of pre- and postsynaptic spikes affects the synaptic weight change. Discuss the implications of synaptic modifications based on different time intervals between spikes.

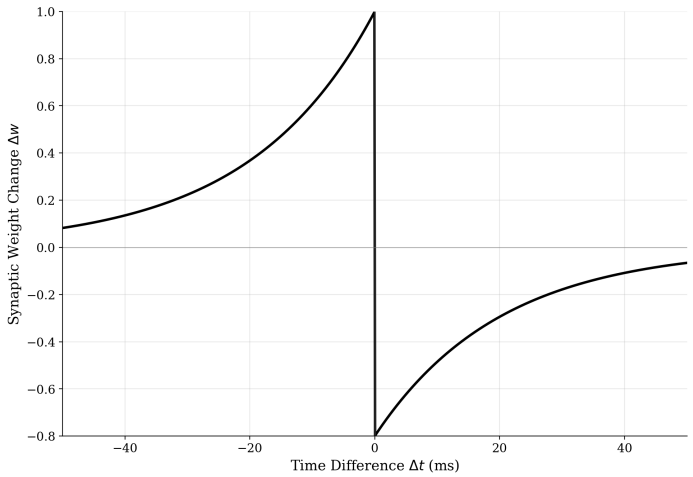

Fig. 2. A diagram illustrating the spike-timing dependent plasticity (STDP) curve. The x-axis represents the time difference between pre- and postsynaptic spikes (Δt), with positive values indicating pre-before-post and negative values indicating post-before-pre. The y-axis represents the change in synaptic weight (Δw). The curve should show potentiation (positive Δw) for pre-before-post and depression (negative Δw) for post-before-pre, with a zero crossing at Δt = 0. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows the opposite relationship between spike timing. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows the exact opposite relationship between. Regenerate using EXACTLY the values and sequences specified.

Question 3

10.0 points

Examine the diagram below of a crossbar array architecture used for storing synaptic weights in neuromorphic computing. Describe how this architecture facilitates efficient weight storage and retrieval. Discuss any challenges related to scaling and variability as depicted in the diagram.

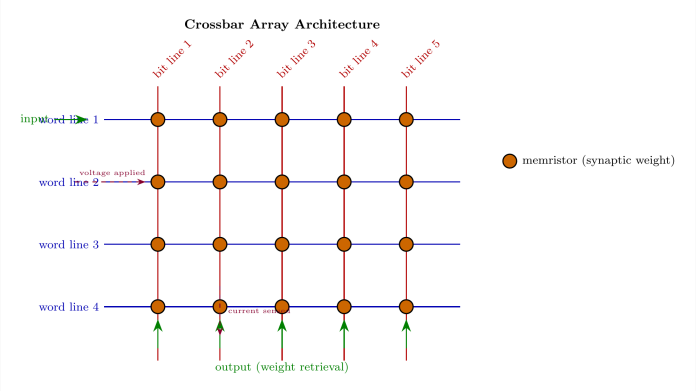

Fig. 3. A crossbar array architecture diagram used in neuromorphic computing. The diagram should show a grid-like structure with horizontal and vertical lines representing rows and columns. Each intersection point (crossbar) should represent a synaptic connection, typically implemented with memristors. Label the rows as 'Word Lines' and the columns as 'Bit Lines'. Include arrows to indicate the flow of data for weight storage and retrieval. Highlight potential challenges such as scaling issues (e.g., increased resistance with more rows/columns) and variability (e.g., differences in memristor characteristics). IMPORTANT: Values or sequences in the diagram don't match the question. The diagram labels for the 'Word Lines'. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows "word line 2" with. Regenerate using EXACTLY the values and sequences specified.

Question 4

10.0 points

Evaluate the temporal coding scheme in the spiking neural network diagram below. Describe how information is encoded temporally and its impact on network processing capabilities. What are the advantages of temporal coding compared to rate coding based on the diagram?

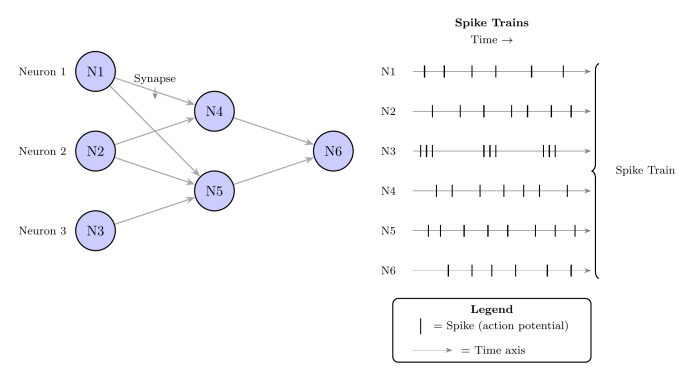

Fig. 4. A spiking neural network diagram showing multiple neurons (nodes) connected by synapses (arrows). Each neuron is labeled, and spike trains (vertical lines) are shown next to each neuron to represent temporal coding. The timing of spikes should vary across neurons to illustrate temporal patterns. Include labels for key components: 'Neuron 1', 'Neuron 2', 'Neuron 3', 'Synapse', 'Spike Train'. The diagram should emphasize the temporal aspect of spike trains, with time represented on the x-axis and spikes as vertical lines. IMPORTANT: Values or sequences in the diagram don't match the question. The labels for the spike trains (N1,. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: The diagram is missing required labels. The diagram is missing the label 'Spike Train. Add each missing label as visible text at the appropriate location in the diagram.

Question 5

10.0 points

Examine the comparative diagram of energy consumption between von Neumann and neuromorphic architectures shown below. Discuss the reasons for differences in energy efficiency. What implications does this have for large-scale neural applications?

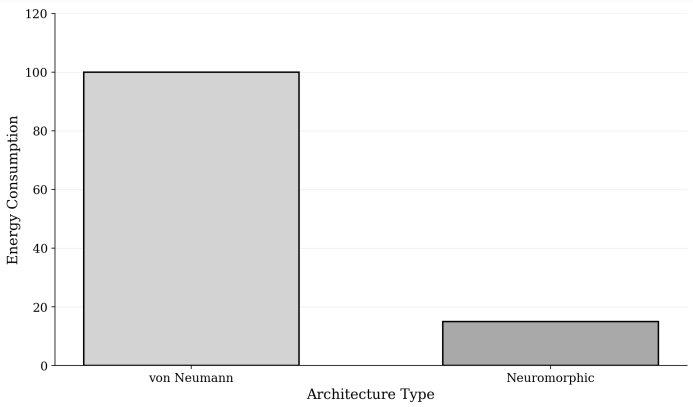

Fig. 5. A comparative bar chart showing the energy consumption of von Neumann and neuromorphic architectures. The x-axis represents the architecture type (von Neumann, neuromorphic), and the y-axis represents energy consumption in arbitrary units. This diagram helps visualize the differences in energy efficiency between the two architectures, aiding in the discussion of their implications for large-scale neural applications.

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